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Term Project
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WigglNET: Niche Market Data Mining and Brokerage Service

Milestone 1: Data Selection and EDA

IMPORTANT NOTE: This project was reframed to validate the effectiveness of the RoBERTa model rather than to predict the accuracy of its outputs. Ground-truth data is needed for prediction but is not immediately available. In practice, it will be collected through client outcomes, feedback, and business results.

Business Question

Small businesses and enterprises across the globe are looking to enhance their AI models with training data. This can be collected through a "Golden Set" created by recently unemployed human reviewers, which is fast but expensive, or through business outcomes and earnings results, which is slower but more reliable. Another way to improve a model is to continuously measure the effectiveness of the model itself, not just its ability to produce the correct output.

For this project, the hypothetical situation is: **a small business needs to improve the effectiveness of a model used to predict the sentiment of entities and keywords in a specific product category**.

A sentiment analysis model called RoBERTa, developed by Facebook, applies labels to keywords and entities based on perceived context. The reference data comes from various social media platforms, like Twitter.

While the RoBERTa model applies sentiment labels to keywords and entities that may be related to the target market, those labels cannot immediately be treated as truth. They are model-generated classifications based on patterns learned during training, not independently verified ground-truth labels.

Without human review, client outcomes, or business results to validate the labels, the project must treat RoBERTa sentiment as a prediction to evaluate rather than an unquestioned fact.

In practice, ground-truth data used to improve RoBERTa would be realized through client outcomes, manual review, or business results. This would happen after the client received the report of keywords and entities labeled by RoBERTa. WigglNET would then work with the client to monitor outcomes based on the decisions informed by RoBERTa, confirming or denying the labels produced by the model.

Therefore, the question being asked in this business problem must be: ***"Can we validate the sentiment labels produced by RoBERTa for a specific product market niche?"*** Doing so would give extra reassurance to the initial pattern-based report provided to clients.

Exploring the Data

A static CSV file contains the results of a query that pulls combined article, keyword, entity, and sentiment information by joining four other tables on either ARTICLE_ID or KEYWORD_ID. Because this is time-series data, the query can ignore all rows where PUBLISHED_DATE is NULL.

It includes the CONFIDENCE field from ARTICLE_ENTITY_SENTIMENT, which indicates the RoBERTa model's confidence level (0-1) in the sentiment classification for each entity.

In [34]:
```
# Imports and Processing
import os
import sys
import warnings
import pandas as pd

os.environ['PYTHONWARNINGS'] = 'ignore'
warnings.filterwarnings('ignore')

df = pd.read_csv('keywords.csv')
```

In [35]:
```
# Display basic info about the dataset
print(f"Dataset shape: {df.shape}")
print(f"\nColumn names: {df.columns.tolist()}")
```

```
Dataset shape: (19367, 9)

Column names: ['ARTICLE_ID', 'TITLE', 'PUBLISHED_DATE', 'KEYWORD_TEXT', 'ENTITY_TEXT', 'ENTITY_TYPE', 'SENTIMENT', 'CONFIDENCE',
'MENTIONS']
```

Predictor Variables

To validate RoBERTa sentiment labels, the model will use the following predictor variables.

- ENTITY_TYPE
- KEYWORD_TEXT
- ENTITY_TEXT
- TITLE
- PUBLISHED_DATE

IMPORTANT NOTE: `ENTITY_TYPE`, `KEYWORD_TEXT`, and `ENTITY_TEXT` are produced by an upstream Named Entity Recognition (NER) model. NER will be evaluated with the same rigor as RoBERTa, but under a separate workstream. The current project is scoped exclusively to RoBERTa.

How many articles are in this dataset?

In [36]:
```
total_articles = df["ARTICLE_ID"].nunique()
print(f"There are {total_articles} articles in this dataset.")
```

```
There are 132 articles in this dataset.
```

If there are only 132 articles, why are there 19,637 rows?

This dataset includes metadata about keywords and entities in an article, not just a list of each article. Every keyword has entity text, an entity type, a sentiment label, and a number of mentions.

What are the max and min published dates?

In [37]:
```
max_pub = df["PUBLISHED_DATE"].max()
print(f"Max published date is {max_pub}")

min_pub = df["PUBLISHED_DATE"].min()
print(f"Min published date is {min_pub}")
```

```
Max published date is 2026-04-10
Min published date is 2022-04-01
```

Identifying Entities with Positive or Negative Sentiment, Ordered by Negative Sentiment

This analysis focuses on entities with strong sentiment signals (positive or negative), excluding purely neutral entities, and orders them by negative sentiment to highlight problematic entities first. A client might use this graph to understand what brands or keywords to avoid in future business decisions.

Ordering by negative mentions (rather than total volume) surfaces entities that would otherwise be hidden under high-volume neutral coverage, making it the more useful view for a brokerage service flagging risk.

Data Preparation

In [38]:
```
# Import visualization libraries
import matplotlib.pyplot as plt
import numpy as np

plt.style.use('ggplot')

# Sentiment color scheme and stack order (reused across charts)
colors = {
    'negative': '#e74c3c',  # Red
    'neutral':  '#95a5a6',  # Gray
    'positive': '#2ecc71'   # Green
}
sentiment_order = ['negative', 'neutral', 'positive']
```

Aggregate by entity and sentiment, then identify entities that have *any* positive or negative sentiment (filtering out purely neutral entities). Sort by negative mention count to surface the highest-risk entities first.

In [39]:
```
entity_sentiment = df.groupby(['ENTITY_TEXT', 'SENTIMENT'])['MENTIONS'].sum().reset_index()

# find entities that have positive OR negative sentiment (not just neutral)
entity_sentiment_set = df.groupby('ENTITY_TEXT')['SENTIMENT'].apply(lambda x: set(x))
                                                              # convert each groups sentiment values to a Python set (removing du

# filter to entities that have 'positive' or 'negative' in their sentiment set
entities_with_sentiment = entity_sentiment_set[
    entity_sentiment_set.apply(lambda x: 'positive' in x or 'negative' in x)
].index.tolist() # convert to list

print(f"Found {len(entities_with_sentiment)} entities with positive or negative sentiment")
```

```
Found 399 entities with positive or negative sentiment
```

In [40]:
```
# Get negative mention counts for these entities
negative_mentions = df[
    (df['ENTITY_TEXT'].isin(entities_with_sentiment)) &
    (df['SENTIMENT'] == 'negative')
].groupby('ENTITY_TEXT')['MENTIONS'].sum()
```

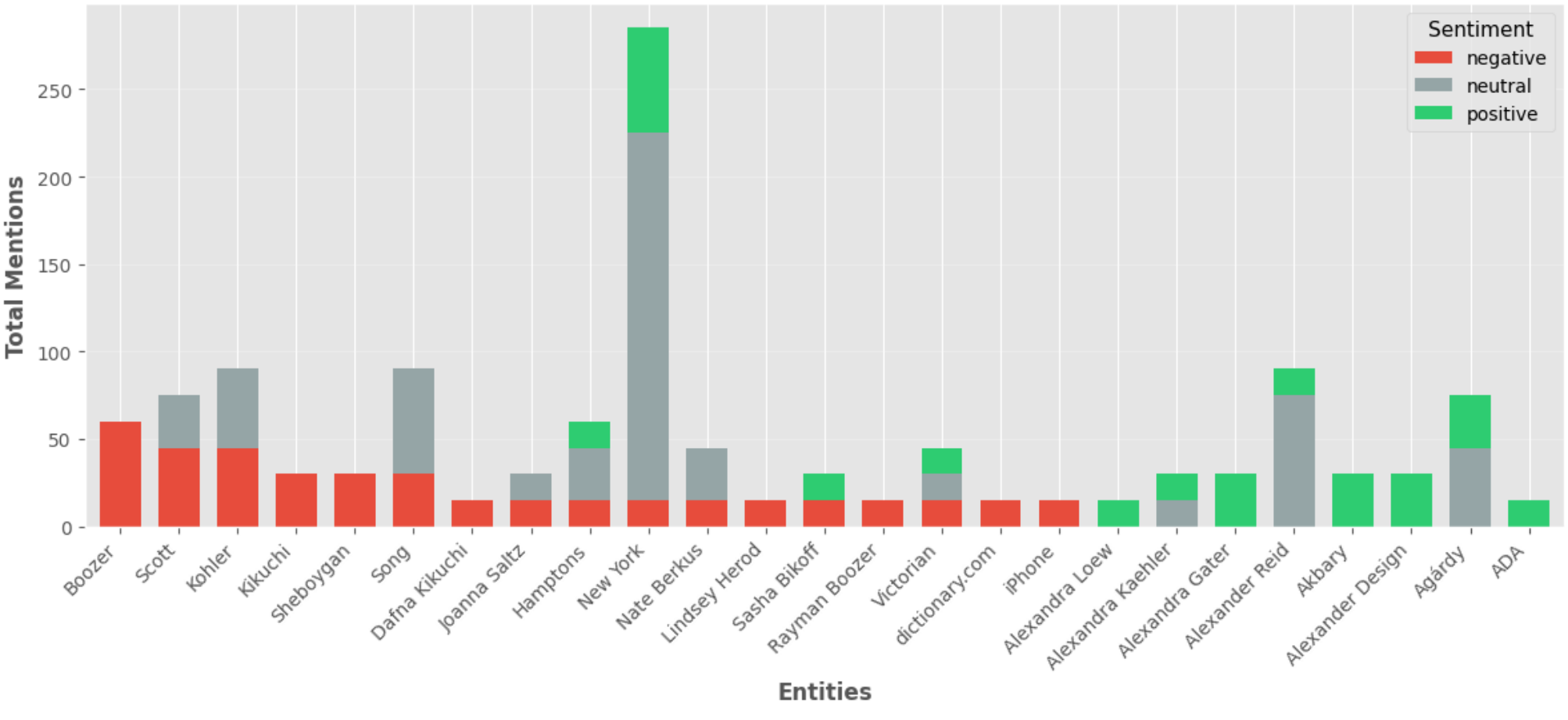
In [41]:
```
# sort by negative mentions (descending) and get top 10
all_sentiment_entities = pd.Series(0, index=entities_with_sentiment)
all_sentiment_entities.update(negative_mentions)
top_entities_by_negative = all_sentiment_entities.nlargest(25).index.tolist()

# filter to only include selected entities
entity_sentiment_top_negative = entity_sentiment[entity_sentiment['ENTITY_TEXT'].isin(top_entities_by_negative)]
```

Visualization

In [42]:
```
# Pivot the data so sentiments become columns (same logic as first viz)
pivot_entity_negative = entity_sentiment_top_negative.pivot(index='ENTITY_TEXT',
                                                            columns='SENTIMENT',
                                                            values='MENTIONS').fillna(0)

# Reorder columns to control the stack order
pivot_entity_negative = pivot_entity_negative.reindex(columns=sentiment_order, fill_value=0)
```

```
# Sort by NEGATIVE mentions (descending) - key difference from first visualization
pivot_entity_negative = pivot_entity_negative.sort_values('negative', ascending=False)
```

In [43]:
```
# create stacked bar chart (same visualization logic as first chart)
fig, ax = plt.subplots(figsize=(12, 6))

pivot_entity_negative.plot(kind='bar',
                           stacked=True,
                           ax=ax,
                           color=[colors[col] for col in pivot_entity_negative.columns],
                           width=0.7)

# Customize the chart
ax.set_title('Top 10 Entities with Positive/Negative Sentiment (Ordered by Negative Mentions)',
             fontsize=16, fontweight='bold', pad=20)
ax.set_xlabel('Entities', fontsize=12, fontweight='bold')
ax.set_ylabel('Total Mentions', fontsize=12, fontweight='bold')
ax.legend(title='Sentiment', title_fontsize=11, fontsize=10)
ax.set_xticklabels(pivot_entity_negative.index, rotation=45, ha='right')

# Add grid for better readability
ax.yaxis.grid(True, alpha=0.3)
ax.set_axisbelow(True)

plt.tight_layout()
plt.show()

print("\nNote: Entities are ordered left-to-right by negative mention count (highest first).")
```




```
Note: Entities are ordered left-to-right by negative mention count (highest first).
```

Observation

This graph brings it all together. The entities with the highest negative sentiment should appear on the far left, decreasing to the right, where entities have more neutral and/or positive sentiment. A client might use this graph to determine which entities to avoid most, while also identifying which entities have the highest negative sentiment with the least positive or neutral sentiment.

This prioritization helps e-commerce clients quickly identify high-risk entities while understanding their full sentiment context. An entity with 100 negative mentions and 200 positive mentions tells a different story than one with 100 negative mentions and 0 positive mentions. Both are shown here, but they are ordered by the strength of the negative signal.

Analyzing Confidence Scores for Negative Sentiment

This pipeline uses RoBERTa, which provides confidence scores (0-1) indicating how certain the model is about each sentiment classification. Examining confidence alongside negative mention counts distinguishes reliable warning signals from potentially misclassified or ambiguous entities.

The same approach can be applied to positive sentiment to surface partnership candidates, but it is omitted here for brevity. The methodology is identical, with the color map and sort field swapped.

Reusable Visualization Functions

In [44]:
```
import matplotlib.cm as cm
from matplotlib.colors import Normalize

def plot_confidence_bar_chart(data, mentions_col, confidence_col,
                              sentiment_type, color_map='Reds'):
    # sort by mentions for better visualization
    data_sorted = data.sort_values(mentions_col, ascending=True)
```

```
# create figure object and ax
fig, ax = plt.subplots(figsize=(12, 10))

# normalize the confidence scores across the data,
#   so that it can be color mapped
norm = Normalize(vmin=data_sorted[confidence_col].min(),
                 vmax=data_sorted[confidence_col].max())
cmap = cm.get_cmap(color_map) # call in color map

# Create bars with color based on confidence
ax.barh(range(len(data_sorted)),
        data_sorted[mentions_col],
        color=cmap(norm(data_sorted[confidence_col])))

# Customize chart
ax.set_yticks(range(len(data_sorted)))
ax.set_yticklabels(data_sorted['ENTITY_TEXT'], fontsize=10)
ax.set_xlabel(f'Total {sentiment_type} Mentions', fontsize=12, fontweight='bold')
ax.set_ylabel('Entity', fontsize=12, fontweight='bold')
ax.set_title(f'Top 25 {sentiment_type}ly Mentioned Entities\n(Color Intensity = Confidence Level)',
             fontsize=14, fontweight='bold', pad=20)

# Add grid
ax.xaxis.grid(True, alpha=0.3)
ax.set_axisbelow(True)

# Add colorbar
sm = cm.ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
cbar = plt.colorbar(sm, ax=ax)
cbar.set_label('Average Confidence Score', fontsize=11, fontweight='bold')

# Add value labels on bars
for i, (mentions, confidence) in enumerate(zip(data_sorted[mentions_col],
                                               data_sorted[confidence_col])):
    ax.text(mentions + 1, i, f'{int(mentions)} ({confidence:.2f})',
            va='center', fontsize=8, color='black')

plt.tight_layout()
plt.show()
```

In [45]:

```
def plot_confidence_scatter(data, mentions_col, confidence_col,
                            sentiment_type, color_map='Reds',
                            quadrant_color_high='#ffcccc', quadrant_color_low='#ffffcc',
                            max_mentions=None):
    # Create figure
    fig, ax = plt.subplots(figsize=(14, 8))

    # Plot scatter points
    scatter = ax.scatter(data[mentions_col],
                         data[confidence_col],
                         s=200,
                         c=data[confidence_col],
                         cmap=color_map,
                         alpha=0.6,
                         edgecolors='black',
                         linewidth=1.5)

    # Add entity labels
    for idx, row in data.iterrows():
        ax.annotate(row['ENTITY_TEXT'],
                    (row[mentions_col], row[confidence_col]),
                    fontsize=9,
                    xytext=(5, 5),
                    textcoords='offset points',
                    bbox=dict(boxstyle='round,pad=0.3', facecolor='yellow', alpha=0.3))

    # Add quadrant lines (median splits)
    median_mentions = data[mentions_col].median()
    median_confidence = data[confidence_col].median()

    ax.axvline(median_mentions, color='gray', linestyle='--', alpha=0.5, linewidth=1)
    ax.axhline(median_confidence, color='gray', linestyle='--', alpha=0.5, linewidth=1)

    # Add quadrant labels
    ax.text(median_mentions * 1.5, data[confidence_col].max() * 0.98,
            'High Confidence\nHigh Volume', ha='center', fontsize=10,
            bbox=dict(boxstyle='round', facecolor=quadrant_color_high, alpha=0.7))
    ax.text(median_mentions * 0.5, data[confidence_col].max() * 0.98,
            'High Confidence\nLow Volume', ha='center', fontsize=10,
            bbox=dict(boxstyle='round', facecolor=quadrant_color_low, alpha=0.7))

    # Customize chart
    ax.set_xlabel(f'Total {sentiment_type} Mentions', fontsize=12, fontweight='bold')
    ax.set_ylabel('Average Confidence Score', fontsize=12, fontweight='bold')
    ax.set_title(f'Confidence vs. {sentiment_type} Mention Count for Top 25 Entities',
                 fontsize=14, fontweight='bold', pad=20)
    ax.grid(True, alpha=0.3)

    # Cap the x-axis if max_mentions is specified
    if max_mentions is not None:
        ax.set_xlim(0, max_mentions)
```

```
    # Add colorbar
    cbar = plt.colorbar(scatter, ax=ax)
    cbar.set_label('Confidence Score', fontsize=11, fontweight='bold')

    plt.tight_layout()
    plt.show()
```

Filter negative sentiment

In [46]:
```
# Filter for negative sentiment only
df_negative = df[df['SENTIMENT'] == 'negative'].copy()

print(f"Found {len(df_negative)} negative sentiment records")
print(f"Confidence score range: {df_negative['CONFIDENCE'].min():.3f} to {df_negative['CONFIDENCE'].max():.3f}")
print(f"Average confidence: {df_negative['CONFIDENCE'].mean():.3f}")
```

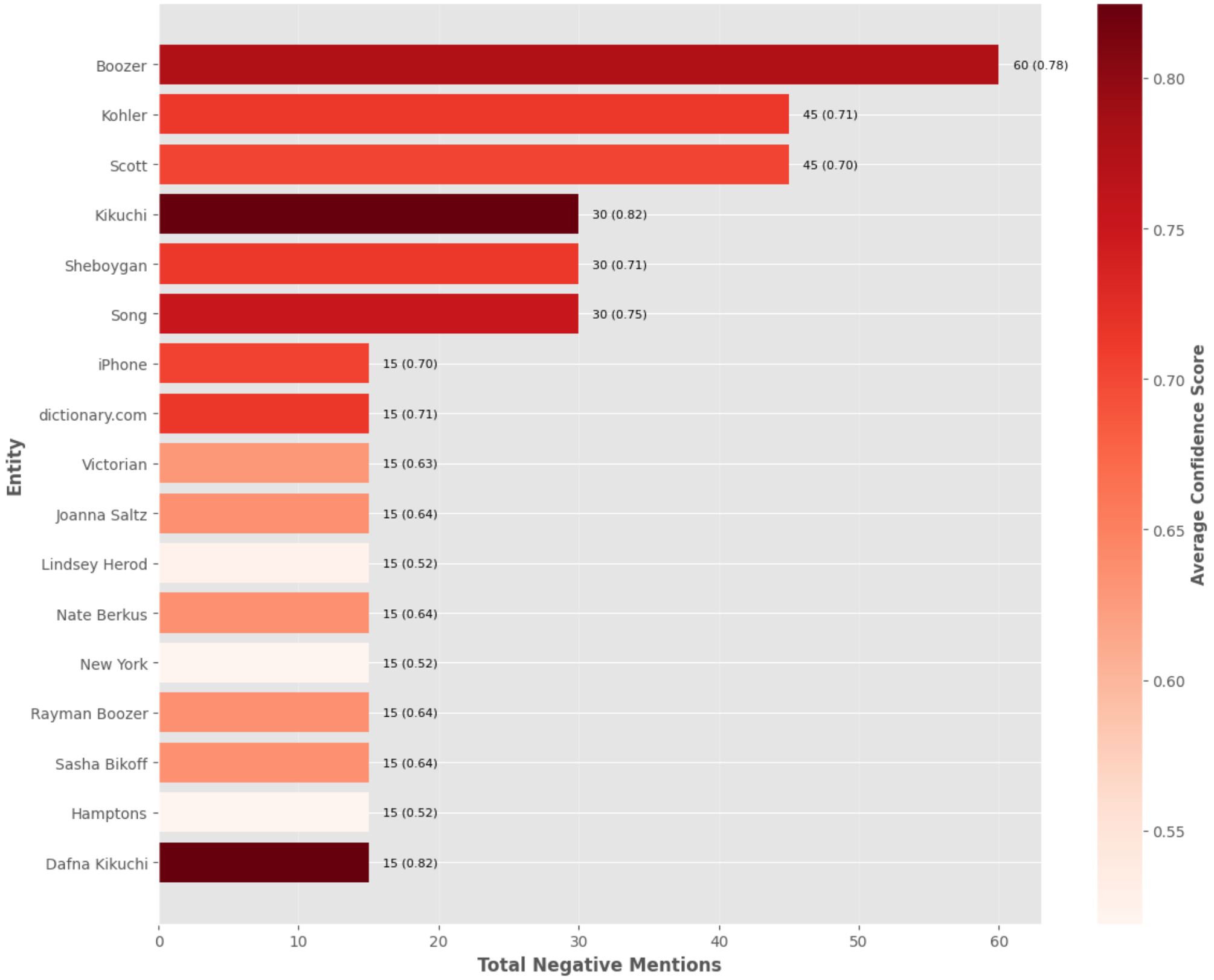
```
Found 255 negative sentiment records
Confidence score range: 0.519 to 0.825
Average confidence: 0.674
```

In [47]:
```
# Aggregate negative entities: sum mentions and average confidence
negative_entity_stats = df_negative.groupby('ENTITY_TEXT').agg({
    'MENTIONS': 'sum',
    'CONFIDENCE': 'mean'
}).reset_index()

negative_entity_stats.columns = ['ENTITY_TEXT', 'TOTAL_NEGATIVE_MENTIONS', 'AVG_CONFIDENCE']
top_25_negative = negative_entity_stats.nlargest(25, 'TOTAL_NEGATIVE_MENTIONS')
top_25_negative['AVG_CONFIDENCE'] = top_25_negative['AVG_CONFIDENCE'].round(3)
```

Bar Chart: Color Gradient by Confidence

Bar height represents negative mention count, while color intensity shows confidence level. Darker bars indicate higher model confidence in the negative sentiment classification.

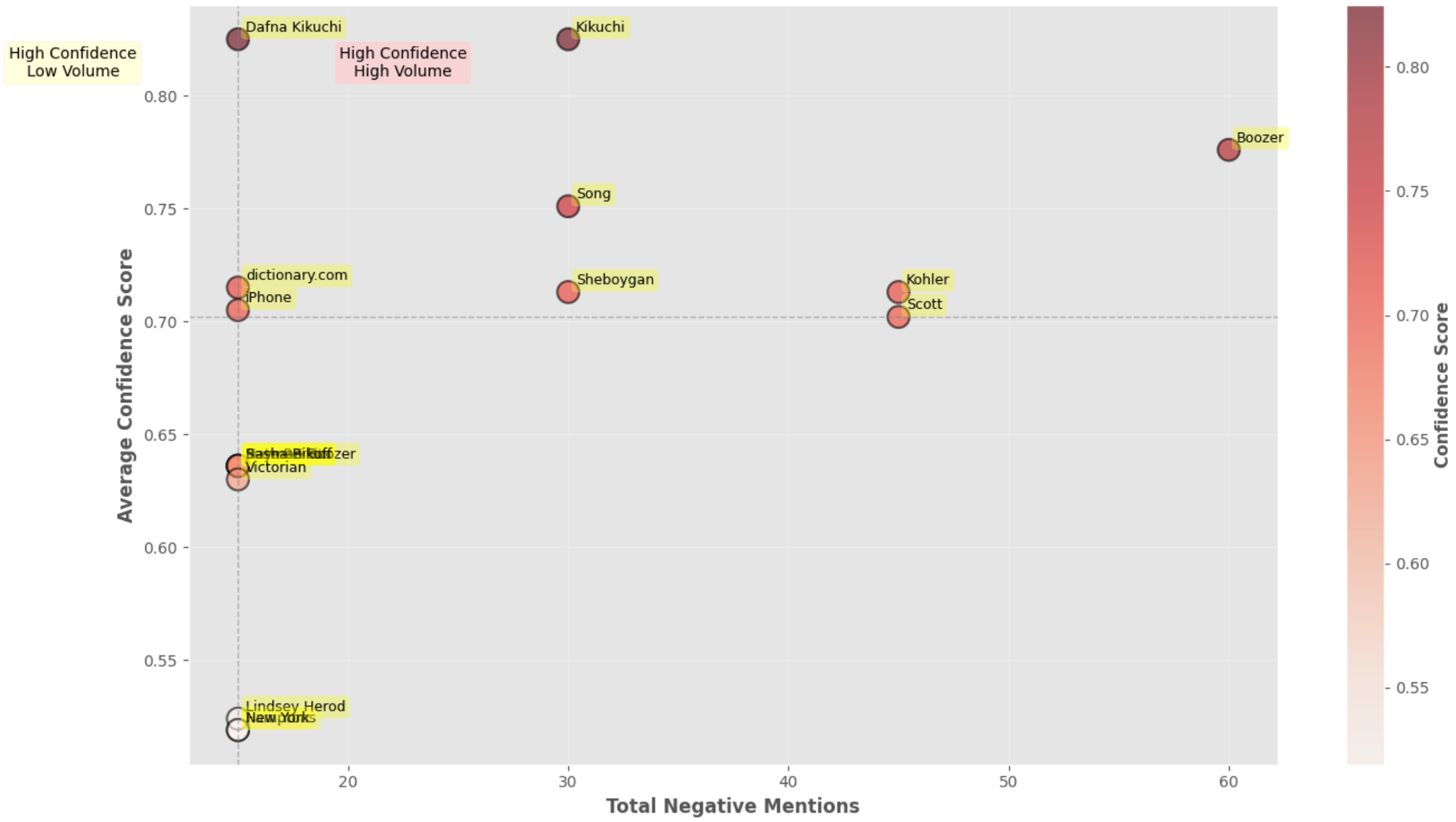
In [48]:
```
# Use the reusable function for negative entities bar chart
plot_confidence_bar_chart(
    data=top_25_negative,
    mentions_col='TOTAL_NEGATIVE_MENTIONS',
    confidence_col='AVG_CONFIDENCE',
    sentiment_type='Negative',
    color_map='Reds'
)
```



Scatter Plot: Confidence vs. Negative Mentions

The top-right quadrant (high mentions, high confidence) represents the most reliable warning signs. High-confidence/low-mention entities are emerging concerns, high-mention/low-confidence entities may be misclassifications worth investigating, and low/low entities are the least concerning.

In [49]:
```
# Use the reusable function for negative entities scatter plot
plot_confidence_scatter(
    data=top_25_negative,
    mentions_col='TOTAL_NEGATIVE_MENTIONS',
    confidence_col='AVG_CONFIDENCE',
    sentiment_type='Negative',
    color_map='Reds',
    quadrant_color_high='#ffcccc',
    quadrant_color_low='#ffffcc'
)
```



As mentioned before, entities on the far right side of the scatter plot have higher mentions, while those in the upper-right area have high confidence scores.

These are what the different confidence thresholds mean:

(>0.85) - the RoBERTa model is very certain about the sentiment classification
(0.70-0.85) - the model is moderately certain, which is typical for clear negative language
(<0.70) - low confidence: uncertain classification. May indicate ambiguous language or misclassification

For WigglNET clients, high-confidence, high-volume entities will appear as darker, taller bars, indicating the most reliable negative signals. E-commerce sellers should investigate these entities thoroughly before associating with them.

This two-dimensional analysis helps clients distinguish between solid negative signals and potentially uncertain classifications, leading to better-informed business decisions.

Milestone 1 Conclusion

The best way for a seller or manufacturer to understand what is happening in a market is to look directly at what is being discussed. Reading articles one by one and judging whether an entity is positive or negative is inefficient compared to the classification models WigglNET runs to identify the sentiment of keywords and entities across hundreds of articles. This is not to say that manually applying sentiment labels is useless. To develop ground-truth labels for a client's particular market needs, there must be human-made truths that the model can compare its predictions to. From there, the model can improve its predictions based on outcomes, results, and KPIs from the client's analytics.

This notebook reviewed a process for validating results produced by RoBERTa to verify the effectiveness of the model's labeling for a particular target market. It was visualized in three stacked bar charts showing different ways clients could use this data to make decisions about topics with positive and negative sentiment.

Milestone 2: Data Preparation

Next in the pipeline is preparing the data for machine learning model building and evaluation. In the following sections, the data will undergo cleaning, extraction, selection, transformation, and feature engineering.

Drop any features that are not useful for your model building and explain why they are not useful.

In [50]:
```
df = df.drop(columns=["CONFIDENCE"])
# ARTICLE_ID is retained here and used as a group key for splitting (see train/test split below)
```

For model development, the only column that needs to be dropped at this stage is `CONFIDENCE`. It is RoBERTa's confidence in the sentiment label it produced. If this field were fed into a classifier validating the accuracy of the labels, the model would look extremely accurate under false pretenses because the feature it relies on is derived from the very label it is trying to validate.

`ARTICLE_ID` is intentionally retained here. Because each article generates roughly 150 rows, a random row-level split would scatter rows from the same article into both the training and test sets. The TF-IDF title features and target-encoded entity/keyword values for those rows would then appear in both partitions, allowing the model to memorize article-level patterns rather than generalize across articles. Keeping `ARTICLE_ID` available until the split step allows a group-based split that places all rows from any given article in either training or test, never both.

Check for missing data

In [51]:
```
# Sum the NaNs
df.isnull().sum()
```

Out[51]:
```
ARTICLE_ID        0
TITLE             0
PUBLISHED_DATE    0
KEYWORD_TEXT      0
ENTITY_TEXT       0
ENTITY_TYPE       0
SENTIMENT         0
MENTIONS          0
dtype: int64
```

A check using `df.isnull().sum()` returned zero null values across all columns. This is expected: the data was queried from a database populated by a clean upstream pipeline, and the SQL query explicitly filtered out rows where PUBLISHED_DATE IS NOT NULL. No imputation or row removal was required.

Engineering new features: Published Date

From the `PUBLISHED_DATE` column, four features will be created:

`PUB_YEAR` : Year the article was published
`PUB_MONTH` : Month the article was published
`PUB_DOW` : Day of the week the article was published

In [52]:
```
# converting to datetime if not already
df['PUBLISHED_DATE'] = pd.to_datetime(df['PUBLISHED_DATE'])

# adding the three new features
df['pub_year']       = df['PUBLISHED_DATE'].dt.year
df['pub_month']      = df['PUBLISHED_DATE'].dt.month
df['pub_dow']        = df['PUBLISHED_DATE'].dt.dayofweek   # 0=Monday, 6=Sunday

# PUBLISHED_DATE is retained here for time-based splitting (see train/test split below)
# It will be excluded from X when separating features from the target
```

Feature transformations: Mentions

In [53]:
```
# Imports and preprocessing
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import LinearSVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, MinMaxScaler, TargetEncoder
```

`MENTIONS` is skewed because a few entities are mentioned hundreds of times. Applying a log transformation compresses the range and helps linear models. The histogram below shows the current distribution of the mentions field, which is right-skewed.

In [54]:
```
# pick a representative spread of values from your data
# Sample representative values from the actual distribution
sample_values = np.unique(
    np.percentile(df['MENTIONS'], [10, 25, 50, 75, 90, 95, 100]).astype(int)
)
log_values = np.log1p(sample_values)

fig, axes = plt.subplots(2, 1, figsize=(14, 5))

# === Top: raw MENTIONS number line ===
axes[0].hlines(0, 0, 40, colors='black', linewidth=1)
axes[0].scatter(sample_values, [0]*len(sample_values),
                color='#e74c3c', s=120, zorder=3, edgecolor='black')

# Label each point with its value
for v in sample_values:
    axes[0].annotate(str(v), (v, 0), textcoords='offset points',
                     xytext=(0, 12), ha='center', fontsize=10)

axes[0].set_xlim(-1, 41)
axes[0].set_ylim(-1, 1)
axes[0].set_title('Before: Raw MENTIONS', fontsize=13, fontweight='bold', loc='left')
axes[0].set_yticks([])
axes[0].spines['top'].set_visible(False)
axes[0].spines['right'].set_visible(False)
axes[0].spines['left'].set_visible(False)

# === Bottom: log-transformed number line ===
axes[1].hlines(0, 0, 4, colors='black', linewidth=1)
axes[1].scatter(log_values, [0]*len(log_values),
                color='#2ecc71', s=120, zorder=3, edgecolor='black')

for raw, log_v in zip(sample_values, log_values):
    axes[1].annotate(f'{log_v:.2f}', (log_v, 0), textcoords='offset points',
                     xytext=(0, 12), ha='center', fontsize=10)
```

```
axes[1].set_xlim(-0.1, 4.1)
axes[1].set_ylim(-1, 1)
axes[1].set_title('After: log1p(MENTIONS)', fontsize=13, fontweight='bold', loc='left')
axes[1].set_yticks([])
axes[1].spines['top'].set_visible(False)
axes[1].spines['right'].set_visible(False)
axes[1].spines['left'].set_visible(False)

plt.suptitle('Redistribute Spacing of MENTIONS',
             fontsize=15, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```

Creating a before-and-after comparison of the logarithm transformation makes it look as if not much changed. However, the transformation compressed the long right tail.

In the raw data, a row with 38 mentions sat 37 units away from a row with 1 mention, a distance the model would treat as enormous. After log1p, that same gap shrinks to about 3 units, putting high- and low-mention rows on a comparable scale. The visual shape does not change dramatically because most rows already cluster at 1-3 mentions, but the underlying distance relationships between values are now proportional, which is what linear models actually need.

In [55]:
```
# Apply the log transformation to MENTIONS before splitting the data
df['MENTIONS'] = np.log1p(df['MENTIONS'])

print(df['MENTIONS'].describe())
```

```
count    19367.000000
mean         1.045827
std          0.474284
min          0.693147
25%          0.693147
50%          0.693147
75%          1.098612
max          3.663562
Name: MENTIONS, dtype: float64
```

Separating the Target from Predictor Variables

To keep the model from accidentally learning from the answer column, `SENTIMENT` is separated into `y`, and the remaining columns are stored in `X` as predictor variables.

In [56]:
```
from sklearn.model_selection import GroupShuffleSplit

# Save group and date keys before excluding them from the feature matrix
groups     = df['ARTICLE_ID']
pub_dates  = df['PUBLISHED_DATE']

X = df.drop(columns=['SENTIMENT', 'ARTICLE_ID', 'PUBLISHED_DATE'])
y = df['SENTIMENT']

print('Predictor shape:', X.shape)
print('Target shape:', y.shape)
print('Target classes:')
print(y.value_counts())
```

```
Predictor shape: (19367, 8)
Target shape: (19367,)
Target classes:
SENTIMENT
neutral     11890
positive     7222
negative      255
Name: count, dtype: int64
```

Performing the Train/Test Split

A group-based split is used instead of a random row-level split. `GroupShuffleSplit` guarantees that every row belonging to a given `ARTICLE_ID` lands entirely in training or entirely in testing, but never both. This prevents the model from seeing the same article's TF-IDF title features and target-encoded entity/keyword values in both partitions, which would allow it to memorize article-level patterns rather than learn generalizable sentiment signals.

An alternative is a **time-based split**: sort articles by `PUBLISHED_DATE` and use the oldest 80% for training and the most recent 20% for testing. This better reflects deployment conditions where the model is trained on historical data and evaluated on future articles. Both approaches are more honest than a random split over raw rows.

In [57]:
```
gss = GroupShuffleSplit(n_splits=1, test_size=0.2, random_state=42)
train_idx, test_idx = next(gss.split(X, y, groups=groups))

X_train = X.iloc[train_idx]
X_test  = X.iloc[test_idx]
y_train = y.iloc[train_idx]
y_test  = y.iloc[test_idx]

train_articles = groups.iloc[train_idx].nunique()
test_articles  = groups.iloc[test_idx].nunique()
overlap        = len(set(groups.iloc[train_idx]) & set(groups.iloc[test_idx]))

print(f'X_train shape:       {X_train.shape}')
print(f'X_test shape:        {X_test.shape}')
print(f'Articles in train:   {train_articles}')
print(f'Articles in test:    {test_articles}')
print(f'Article overlap:     {overlap}  (must be 0)')
```

```
X_train shape:       (15317, 8)
X_test shape:        (4050, 8)
Articles in train:   105
Articles in test:    27
Article overlap:     0  (must be 0)
```

Target Encoding: Text and Keywords

`ENTITY_TEXT` and `KEYWORD_TEXT` cannot be read by a machine learning model as text. Target encoding converts each category to a numeric value based on the target values in the training data. To avoid data leakage, the encoder is fit only on `X_train` and `y_train`, then applied to both `X_train` and `X_test`.

In [58]:
```
sentiment_map = {
    'negative': 0,
    'neutral': 1,
    'positive': 2
}

y_train_encoded = y_train.map(sentiment_map)

target_encoder = TargetEncoder(target_type='continuous', random_state=42)

train_target_encoded = target_encoder.fit_transform(
    X_train[['ENTITY_TEXT', 'KEYWORD_TEXT']],
    y_train_encoded
)

test_target_encoded = target_encoder.transform(
    X_test[['ENTITY_TEXT', 'KEYWORD_TEXT']]
)

train_target_encoded = pd.DataFrame(
    train_target_encoded,
    columns=['ENTITY_ENC', 'KEYWORD_ENC'],
    index=X_train.index
)

test_target_encoded = pd.DataFrame(
    test_target_encoded,
    columns=['ENTITY_ENC', 'KEYWORD_ENC'],
    index=X_test.index
)

X_train = X_train.drop(columns=['ENTITY_TEXT', 'KEYWORD_TEXT']).join(train_target_encoded)
X_test = X_test.drop(columns=['ENTITY_TEXT', 'KEYWORD_TEXT']).join(test_target_encoded)

print('After target encoding:')
print(X_train[['ENTITY_ENC', 'KEYWORD_ENC']].describe())
```

```
After target encoding:
         ENTITY_ENC   KEYWORD_ENC
count  15317.000000  15317.000000
mean       1.356151      1.355138
std        0.453562      0.152326
min        0.000000      0.340672
25%        1.000000      1.302734
50%        1.000000      1.358929
75%        2.000000      1.400606
max        2.000000      2.000000
```

TF-IDF Encoding: Title

`TfidfVectorizer` is fit on the training titles only. The fitted vectorizer is then used to transform both the training and test titles so the same title vocabulary is used in both datasets.

In [59]:
```
tfidf = TfidfVectorizer(
    max_features=100,
    stop_words='english',
    ngram_range=(1, 2)
```

```
)

train_title_matrix = tfidf.fit_transform(X_train['TITLE'])
test_title_matrix = tfidf.transform(X_test['TITLE'])

title_columns = [f'title_{w}' for w in tfidf.get_feature_names_out()]

train_title_df = pd.DataFrame(
    train_title_matrix.toarray(),
    columns=title_columns,
    index=X_train.index
)

test_title_df = pd.DataFrame(
    test_title_matrix.toarray(),
    columns=title_columns,
    index=X_test.index
)

X_train = X_train.drop(columns=['TITLE']).join(train_title_df)
X_test = X_test.drop(columns=['TITLE']).join(test_title_df)

print(f'After TF-IDF, added {train_title_df.shape[1]} title-derived columns')
print('Sample title features:', title_columns[:10])
```

```
After TF-IDF, added 100 title-derived columns
Sample title features: ['title_2026', 'title_2026 color', 'title_25', 'title_45', 'title_alyssa', 'title_alyssa khan', 'title_an
nounces', 'title_announces 2026', 'title_bathroom', 'title_bathroom vanity']
```

One Hot Encoding: Entity Type

`ENTITY_TYPE` has no meaningful ranking, so one-hot encoding converts it to dummy variables. The test set is aligned to the training dummy columns so both datasets have the same feature layout.

In [60]:
```
train_entity_type_dummies = pd.get_dummies(X_train['ENTITY_TYPE'], prefix='etype', dtype=int)
test_entity_type_dummies = pd.get_dummies(X_test['ENTITY_TYPE'], prefix='etype', dtype=int)

test_entity_type_dummies = test_entity_type_dummies.reindex(
    columns=train_entity_type_dummies.columns,
    fill_value=0
)

X_train = X_train.drop(columns=['ENTITY_TYPE']).join(train_entity_type_dummies)
X_test = X_test.drop(columns=['ENTITY_TYPE']).join(test_entity_type_dummies)

print('After one-hot encoding ENTITY_TYPE:')
print(X_train.filter(like='etype_').head())
```

```
After one-hot encoding ENTITY_TYPE:
   etype_GPE  etype_NORP  etype_ORG  etype_PERSON  etype_PRODUCT
0          0           0          1             0              0
1          0           0          1             0              0
2          0           0          0             1              0
3          0           0          0             1              0
4          0           1          0             0              0
```

Ready for Modeling

All feature columns are now numeric and were prepared using the training data first. `SENTIMENT` remains separate as the target variable.

In [61]:
```
print('Training feature types:')
print(X_train.dtypes.value_counts())

print('Testing feature types:')
print(X_test.dtypes.value_counts())
```

```
Training feature types:
float64    103
int64        5
int32        3
Name: count, dtype: int64
Testing feature types:
float64    103
int64        5
int32        3
Name: count, dtype: int64
```

Quick Overview of the Data

Below are the final results of the training data after model preparation.

In [62]:
```
X_train
```

Out[62]:

	MENTIONS	pub_year	pub_month	pub_dow	ENTITY_ENC	KEYWORD_ENC	title_2026	title_2026 color	title_25	title_45	...	title_vs	title_walk	ti
0	1.386294	2025	11	2	1.000000	1.400606	0.0	0.0	0.0	0.395918	...	0.0	0.0	
1	1.386294	2025	11	2	1.410054	1.398286	0.0	0.0	0.0	0.395918	...	0.0	0.0	
2	1.098612	2025	11	2	1.000000	1.406669	0.0	0.0	0.0	0.395918	...	0.0	0.0	
3	1.098612	2025	11	2	2.000000	1.400606	0.0	0.0	0.0	0.395918	...	0.0	0.0	
4	1.098612	2025	11	2	1.574908	1.398286	0.0	0.0	0.0	0.395918	...	0.0	0.0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
19362	1.098612	2024	3	1	2.000000	1.292936	0.0	0.0	0.0	0.000000	...	0.0	0.0	
19363	1.098612	2024	3	1	1.000000	1.258823	0.0	0.0	0.0	0.000000	...	0.0	0.0	
19364	0.693147	2024	3	1	2.000000	1.231292	0.0	0.0	0.0	0.000000	...	0.0	0.0	
19365	0.693147	2024	3	1	1.000000	1.419574	0.0	0.0	0.0	0.000000	...	0.0	0.0	
19366	0.693147	2024	3	1	1.000000	1.419574	0.0	0.0	0.0	0.000000	...	0.0	0.0	

15317 rows × 111 columns

Milestone 3: Model Building and Evaluation

Classification Model: Logistic Regression

Logistic regression is used as a baseline classification model. The model is trained only on `X_train` and `y_train`, then evaluated on the unseen `X_test` rows.

In [63]:
```
from sklearn.metrics import f1_score, recall_score

log_reg_model = Pipeline(steps=[
    ('scaler', StandardScaler()),
    ('classifier', LogisticRegression(max_iter=2000))
])

log_reg_model.fit(X_train, y_train)
y_pred = log_reg_model.predict(X_test)

macro_f1    = f1_score(y_test, y_pred, average='macro')
neg_recall  = recall_score(y_test, y_pred, labels=['negative'], average=None)[0]

print(f'Accuracy:              {accuracy_score(y_test, y_pred):.4f}')
print(f'Macro F1:              {macro_f1:.4f}')
print(f'Negative-class recall: {neg_recall:.4f}')
print()
print('Classification report:')
print(classification_report(y_test, y_pred))
print('Confusion matrix:')
print(confusion_matrix(y_test, y_pred))
```

```
Accuracy:              0.5709
Macro F1:              0.3300
Negative-class recall: 0.0000

Classification report:
              precision    recall  f1-score   support

    negative       0.00      0.00      0.00         0
     neutral       0.63      0.77      0.69      2535
    positive       0.40      0.24      0.30      1515

    accuracy                           0.57      4050
   macro avg       0.34      0.34      0.33      4050
weighted avg       0.54      0.57      0.54      4050

Confusion matrix:
[[   0    0    0]
 [  45 1953  537]
 [   0 1156  359]]
```

The baseline logistic regression model is evaluated here to establish a reference point. Accuracy, macro F1, and negative-class recall are all reported. Macro F1 and negative-class recall are the primary metrics: with only ~1% of rows labeled negative, high accuracy is achievable by largely ignoring that class, so accuracy alone is not a reliable measure of usefulness. The model predicted the RoBERTa-generated sentiment labels correctly — but those labels are themselves model outputs, not ground truth, so these numbers measure RoBERTa-replication ability, not real-world sentiment accuracy.

GridSearchCV with Pipeline Classifiers

After establishing logistic regression as a baseline, `GridSearchCV` can compare multiple classifier algorithms. Each model is placed inside a `Pipeline`, which keeps preprocessing and model training organized. The grid search is fit only on `X_train` and `y_train`; the held-out `X_test` set is used only after the best model is selected.

The step below defines each model's settings before model selection occurs. `StandardScaler` is applied to models that benefit from standardization, while `MinMaxScaler` is used for KNN because KNN is distance-based and can be sensitive to feature scale. Hyperparameters are set for each model so `GridSearchCV` can select the strongest configuration.

In [64]:
```
# The candidate hyperparameter values are set before training,
# and GridSearchCV selects the best-performing values through cross-validation.
model_configs = {
    # Set the baseline model
    'Logistic Regression': {
        'pipeline': Pipeline(steps=[
            ('scaler', StandardScaler()),  # StandardScaler used for baseline model
            ('classifier', LogisticRegression(max_iter=2000))
        ]),
        'params': {  # GridSearchCV tests these combinations
            'classifier__C': np.logspace(-2, 2, 5),
            'classifier__penalty': ['l2']
        }
    },

    # Compare with KNN
    # KNN uses distances between observations to classify new records.
    # Because distance-based models are sensitive to feature scale,
    # MinMaxScaler places numeric predictors on a common 0-to-1 scale
    # before GridSearchCV searches for the best K value.
    'KNN': {
        'pipeline': Pipeline(steps=[
            ('scaler', MinMaxScaler()),
            ('classifier', KNeighborsClassifier())
        ]),
        'params': {  # GridSearchCV tests these combinations
            'classifier__n_neighbors': [3, 5, 7, 9, 11, 15, 21],
            'classifier__weights': ['uniform', 'distance']
        }
    },

    # Compare with Random Forest
    # Random Forest classifiers do not require feature scaling because
    # decision trees make splits based on feature thresholds instead of
    # distance calculations.
    'Random Forest': {
        'pipeline': Pipeline(steps=[
            ('classifier', RandomForestClassifier(random_state=42, n_jobs=-1))
        ]),
        'params': {  # GridSearchCV tests these combinations
            'classifier__n_estimators': [100, 200],
            'classifier__max_depth': [None, 10, 20]
        }
    },

    # Compare with support vector machine
    # SVM is sensitive to feature scaling, so StandardScaler is used to
    # place features on a comparable mean/std scale.
    'Linear SVM': {
        'pipeline': Pipeline(steps=[
            ('scaler', StandardScaler()),
            ('classifier', LinearSVC(max_iter=5000, random_state=42))
        ]),
        'params': {  # GridSearchCV tests these combinations
            'classifier__C': [0.1, 1, 10]
        }
    }
}
```

Model Selection Results

A list of each model's best weighted CV scores is generated to determine which model performed best. This process may take about two to five minutes, depending on processing power.

In [65]:
```
import warnings
from sklearn.exceptions import ConvergenceWarning

warnings.filterwarnings('ignore', category=FutureWarning)
warnings.filterwarnings('ignore', category=ConvergenceWarning)
warnings.filterwarnings('ignore', message='.*sklearn.utils.parallel.delayed.*', category=UserWarning)

grid_results = []
best_estimators = {}

# For each model in the configuration it was given,
for model_name, config in model_configs.items():
    # f1_macro weights all three classes equally, giving the minority negative class
    # the same influence as neutral and positive during model selection.
    grid_search = GridSearchCV(
        estimator=config['pipeline'],
        param_grid=config['params'],
        cv=3,
        scoring='f1_macro',
        n_jobs=-1
    )

    grid_search.fit(X_train, y_train)

    grid_results.append({
        'Model': model_name,
        'Best CV Macro F1': grid_search.best_score_,
```

```
        'Best Parameters': grid_search.best_params_
    })

    best_estimators[model_name] = grid_search.best_estimator_

grid_results_df = pd.DataFrame(grid_results).sort_values(
    by='Best CV Macro F1',
    ascending=False
)

grid_results_df
```

Out[65]:

	Model	Best CV Macro F1	Best Parameters
2	Random Forest	0.740709	{'classifier__max_depth': 20, 'classifier__n_e...
0	Logistic Regression	0.660446	{'classifier__C': 0.01, 'classifier__penalty':...
3	Linear SVM	0.582669	{'classifier__C': 1}
1	KNN	0.503458	{'classifier__n_neighbors': 11, 'classifier__w...

Evaluate the Best Grid Search Model

The model with the highest cross-validation score is selected from the grid search results, then evaluated once on `X_test`.

In [66]:

```
best_model_name = grid_results_df.iloc[0]['Model']
best_grid_model = best_estimators[best_model_name]

y_pred_grid = best_grid_model.predict(X_test)

macro_f1_grid   = f1_score(y_test, y_pred_grid, average='macro')
neg_recall_grid = recall_score(y_test, y_pred_grid, labels=['negative'], average=None)[0]

print('Best model:', best_model_name)
print(f'Accuracy:              {accuracy_score(y_test, y_pred_grid):.4f}')
print(f'Macro F1:              {macro_f1_grid:.4f}')
print(f'Negative-class recall: {neg_recall_grid:.4f}')
print()
print('Classification report:')
print(classification_report(y_test, y_pred_grid))
print('Confusion matrix:')
print(confusion_matrix(y_test, y_pred_grid))
```

```
Best model: Random Forest
Accuracy:              0.6151
Macro F1:              0.4805
Negative-class recall: 0.0000

Classification report:
              precision    recall  f1-score   support

     neutral       0.64      0.90      0.74      2535
    positive       0.45      0.14      0.22      1515

    accuracy                           0.62      4050
   macro avg       0.55      0.52      0.48      4050
weighted avg       0.57      0.62      0.55      4050

Confusion matrix:
[[2276  259]
 [1300  215]]
```

The best model selected by macro F1 cross-validation score is evaluated on the held-out test articles. Because the group-based split ensures no article's rows appear in both training and test, these scores reflect genuine generalization across unseen articles rather than memorization of article-level patterns. Macro F1 and negative-class recall are the headline metrics; accuracy is included for reference but should not be the sole basis for drawing conclusions about model quality given the class imbalance.

With the group-based split in place, the reported scores reflect how well each model generalizes to articles it has never seen during training. Any large drop from the earlier random-split scores is expected and honest; the previous near-perfect accuracy was largely an artifact of the model seeing the same article's rows in both training and test. The macro F1 and negative-class recall figures from the group split are the meaningful benchmarks to carry forward.

These results represent the accuracy of the cross-validation for a model measuring RoBERTa's ability to **replicate labels** - not the validation of the labels themselves.

Final Project Submission Summary

Based on feedback from Milestone three, the following edits have been made to this notebook. To address row leakage, ARTICLE_ID and PUBLISHED_DATE are now kept until after splitting. `GroupShuffleSplit` ensures all rows from any article land in train OR test, but never both. To separate what the model will train from its testing set, group split was implemented. This ensures every row belonging to a given `ARTICLE_ID` lands entirely in training or entirely in testing, but never both. For transparency, the Macro F1 and Negative recall are printed in every eval block. Additionally, GridSearchCV now scores on f1_macro instead of f1_weighted.

As one might expect, the accuracy dropped significantlly after solving for leakage and performing a proper train/test split for the data. Accuracy is now at 65.51%, but this the most honest result after having made the above adjustments. F1 and Recall matter a lot more in a situation like this where a model struggles with the minority class.